

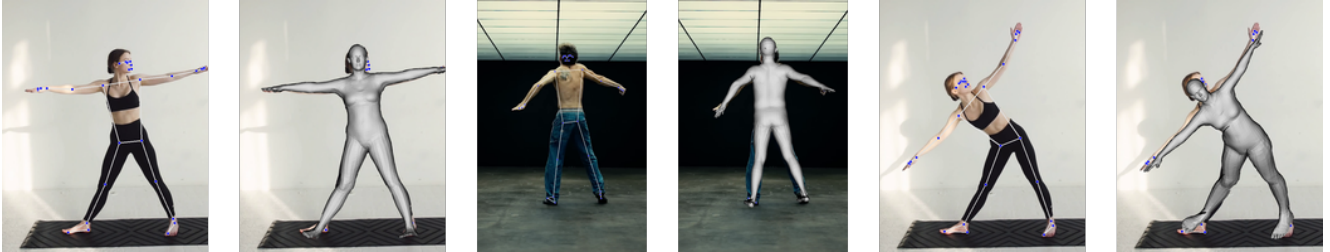
3D Scanning and Motion Capture Body Animation

Nanxing Deng

Gonzalo Martinez Piedrola

Jonathan Herwerth

Elisa Schaerer



This paper presents body animation using RGB video, OpenPose, and the SMPL parametric body model with mesh, skeleton, and skinning.

Abstract

This work presents a C++ optimization-based pipeline that fits the SMPL parametric body model to 2D keypoints detected in each video frame, using the Ceres Solver for nonlinear least squares optimization. We explore single-frame fitting variants like pose-only and joint pose-shape estimation with and without a Gaussian Mixture Model (GMM) pose prior, as well as a prototype multi-frame approach incorporating temporal smoothness. Our results highlight trade-offs between accuracy, runtime, and temporal consistency: single-frame pose-shape fitting yields the lowest reprojection errors, while the tested multi-frame prototype is slower and less reliable within project constraints. Future work should focus on improved temporal regularization, robust initialization, and real-time extensions.

1. Introduction

The task of estimating 3D human pose from videos has gained increasing relevance in applications such as animation, motion analysis, and human-computer interaction. This work aims to implement a C++-based pipeline that inputs a monocular video and outputs a new video in which a 3D SMPL mesh is fitted and visualized for each frame. The core component is an optimization loop that fits the SMPL model parameters (pose, shape, and global translation) to detected 2D key-points in each frame. This is achieved using the Ceres Solver, a C++ library for nonlinear least-squares optimization, in combination with the

Levenberg–Marquardt algorithm. The pipeline focuses on optimization-based fitting to provide high controllability and interpretability, and it can be extended to multi-frame scenarios to improve temporal consistency.

2. Related Work

Modern 3D body fitting relies on the SMPL family of parametric human models. SMPL (Skinned Multi-Person Linear Model) [5] represents the shape and articulation of the human body using a learned statistical model. It forms the backbone of many pose estimation pipelines, with extensions such as SMPL-X [6] that capture more expressive details, including the hands and face. These models serve as strong priors: their shape space captures typical human proportions, and they enforce realistic joint limits and prevent self-intersections, which is crucial during optimization.

Optimization-based methods fit SMPL parameters to 2D joint detections by minimizing the reprojection error between model joints and observed keypoints. The objective is typically augmented with pose and shape priors, and may include penalties to avoid interpenetration [2]. These non-convex problems are commonly solved using gradient-based solvers such as the Ceres Solver [1], a C++ library for large-scale least-squares optimization.

In contrast, learning-based approaches [3, 4] use neural networks to perform end-to-end regression of SMPL parameters directly from images or videos, enabling real-time inference but at the cost of reduced controllability.

In video-based scenarios, a common strategy is to enforce temporal consistency by keeping shape parameters

fixed across frames while allowing pose to vary. Additional smoothness terms can be introduced to encourage natural motion and avoid jitter.

3. Method

3.1. Overview of the Pipeline

We fit a parametric SMPL body to monocular RGB video by minimizing 2D reprojection error between detected image keypoints and the projected 3D model joints. For each frame, we detect 2D landmarks with MediaPipe Pose (33 landmarks with visibility scores), map a SMPL-consistent subset to joints, and optimize pose, shape, and a global similarity transform. We use the Eigen-based `avatar` library for SMPL loading and forward kinematics but implement our own Ceres-based optimization for RGB-only fitting. The final mesh is rendered back onto the frame for qualitative inspection.

3.2. 2D Keypoints and Camera

MediaPipe outputs normalized coordinates and visibility. We discard points with visibility < 0.5 , convert normalized positions to pixels, and form pelvis and chest proxies as hip and shoulder midpoints. Camera intrinsics follow a pinhole model with centered principal point and $f_x = f_y = 0.9 \cdot \max(W, H)$ from the image size. No calibration is estimated; instead, global scale and translation absorb subject distance.

3.3. Model Integration

We evaluated three C++ SMPL implementations. **SM-PLpp** [9] relies on LibTorch, which prevents clean Ceres autodiff integration. **smplxpp** [8] is templated and autodiff-friendly in principle, but practical build and interface frictions blocked timely integration. **avatar** [7] provides Eigen-based model loading, shape blendshapes, joint regression, FK, and a GMM pose prior; we reuse these components only and write custom residuals for our solver.

3.4. Optimization

We estimate SMPL pose $\theta \in \mathbb{R}^{72}$, shape $\beta \in \mathbb{R}^{10}$, and a global similarity $(s, \mathbf{R}, \mathbf{t})$ by minimizing

$$E(\theta, \beta, s, \mathbf{R}, \mathbf{t}) = E_{\text{reproj}} + \lambda_{\text{pose}} E_{\text{pose}} + \lambda_{\text{shape}} E_{\text{shape}}, \quad (1)$$

with

$$E_{\text{reproj}} = \sum_{i \in \mathcal{J}} \rho \left(\left\| \pi(s \mathbf{R} \mathbf{J}_i(\theta, \beta) + \mathbf{t}) - \begin{bmatrix} u_i \\ v_i \end{bmatrix} \right\|^2 \right), \quad (2)$$

$$E_{\text{pose}} = \|\theta\|^2 \quad (\text{or Avatar GMM prior}), \quad (3)$$

$$E_{\text{shape}} = \|\beta\|^2. \quad (4)$$

Here $\mathbf{J}_i(\cdot)$ is the 3D joint via forward kinematics and $\pi([X, Y, Z]^\top) = [f_x X/Z + c_x, f_y Y/Z + c_y]^\top$. The robustifier ρ is the Huber loss with $\delta = 3$ px.

Implementation uses Ceres as a single per-frame non-linear least squares with parameter blocks for the global similarity ($s \in \mathbb{R}$, root angle-axis, translation), all non-root joint angle-axes (23×3), and optionally the 10D shape vector. Each observed joint contributes one 2D reprojection residual with the Huber loss above. A pose prior is applied either as simple L2 or the Avatar GMM prior, and shape uses an L2 prior. We bound the global scale to $s \in [0.3, 3.0]$ and fix clearly unobserved extremities to avoid drift. The camera mapping applies a fixed initial orientation R_0 (yaw 180° with Y -flip), then the optimized root rotation, global scale, and translation, followed by perspective projection. Residuals are implemented with `ceres::DynamicAutoDiffCostFunction`; Ceres uses Levenberg–Marquardt with a dense-QR solver and typically converges in 50–100 iterations per frame.

3.4.1 Ceres-Based Differentiable Kinematics

Our optimization framework implements a custom forward kinematics pipeline specifically designed for efficient gradient computation within the Ceres Solver [1] autodifferentiation framework. The core methodology centers around two templated cost functors: `ReprojCost` for pose-only optimization and `ReprojCostShape` for joint pose-shape estimation.

The forward kinematics computation begins by establishing a reference skeleton configuration through computation of base joint positions in zero-pose, zero-shape configuration. For each joint j , we compute bone offset vectors $\mathbf{o}_j = \mathbf{J}_j^{\text{base}} - \mathbf{J}_{\text{parent}(j)}^{\text{base}}$, where all joint positions are translated such that the root joint lies at the origin for numerical stability.

During optimization, joint positions are computed via explicit kinematic chain traversal: starting from the target joint’s base offset, we iteratively walk up the kinematic tree, applying parent rotations using Ceres’ `AngleAxisRotatePoint` function and accumulating bone translations. This approach avoids expensive matrix chain multiplications while maintaining full compatibility with automatic differentiation. Joint rest positions depend linearly on shape $\beta \in \mathbb{R}^S$ via a joint-shape regressor $A \in \mathbb{R}^{3J \times S}$. For joint j (block $A_j \in \mathbb{R}^{3 \times S}$),

$$J_j(\beta) = J_j^{(0)} + A_j \beta, \quad (5)$$

$$o_j(\beta) = J_j(\beta) - J_{\text{par}(j)}(\beta) \quad (6)$$

$$= o_j^{(0)} + (A_j - A_{\text{par}(j)}) \beta, \quad (7)$$

where $o_j^{(0)} = J_j^{(0)} - J_{\text{par}(j)}^{(0)}$. During FK we add the *relative*

shape correction

$$\Delta o_j(\beta) = \Delta J_j(\beta) - \Delta J_{\text{par}(j)}(\beta) \quad (8)$$

$$= (A_j - A_{\text{par}(j)})\beta, \quad (9)$$

rather than ΔJ_j alone, so shape is applied once per bone (no double-counting along the chain). Without subtracting the parent’s shape shift, the parent’s deformation would be re-added at every descendant when walking up the chain, which distorts bone vectors.

3.5. Single-Frame Fitting and Variants

We support two modes. In *pose-only*, β is fixed to the mean shape; this is faster and robust when detections are sparse. In *joint pose-shape*, β is optimized with an L2 shape prior to personalize limb proportions; this improves alignment but increases nonconvexity and can encourage shape-compensation.

3.6. Multi-Frame Fitting (Prototype)

To exploit temporal context beyond single-frame fitting, we implemented a sliding-window bundle adjustment procedure that jointly optimizes poses for multiple frames while enforcing shared shape and temporal smoothness. The SMPL shape parameters $\mathbf{w} \in \mathbb{R}^S$ are treated as a single global parameter block shared across all frames in the optimization window, ensuring a consistent body shape estimate. In Stage 1, shape and poses are optimized jointly on an anchor subset of frames. In Stage 2, the shape is effectively locked (very high λ_{shape}) while poses are refined in overlapping windows of length W and overlap O , moving forward by $W - O$ frames at a time.

First-order smoothness constraints are added between consecutive frames on root translation, root rotation (angle-axis), and each joint’s angle-axis parameters. For each 3-vector $\mathbf{x}$ from these sets, a residual of the form $r = \lambda_T(\mathbf{x}_f - \mathbf{x}_{f+1})$ is added, weighted by the temporal regularization parameter λ_T (we used ≈ 3.0). This reduces frame-to-frame jitter and helps in cases of partial occlusion by coupling adjacent frames.

The same reprojection cost and pose prior as described in the single-frame fitting section are applied here, but now for all frames in the optimization window, and jointly over the shared shape parameter block. Optimization is performed with Ceres Solver using a dense QR solver, up to `max_iters` iterations and 8 threads.

Due to runtime constraints and engineering trade-offs, this prototype was not tuned to consistently outperform the single-frame fitting baseline, but it demonstrates that temporal coupling and shape sharing can reduce variation and improve robustness in challenging frames.

4. Results

Variant	Mean Pixel Err.	Opt. Time (s)
SF no shape GMM	35.98	0.86
SF shape GMM	28.98	5.24
SF no shape	21.14	0.91
SF shape	20.49	4.38
MF	43.18	6.00

Table 1. Per-frame comparison of variants by mean pixel error and optimization time.

All quantitative experiments ran on a Google Compute Engine VM with Debian GNU/Linux (kernel 6.1.0-37-cloud-amd64), x86_64, and 8 CPU cores. We used Ceres Solver 1.14. Quantitative results report optimization time per frame and mean pixel error per frame. The input for all runs was monocular 2D keypoints and per frame RGB with simple pinhole intrinsics where $f_x = f_y = 0.9 \max(W, H)$ and the principal point is at $(W/2, H/2)$.

We evaluated multiple short videos with a single centered subject, spanning motions from near T pose dances to yoga like poses with self occlusions. This setting stresses monocular depth and scale ambiguity because many 3D poses project almost identically and it also exposes sensitivity to 2D detection quality under motion blur and occlusions. Priors are therefore important since without them joints can drift along the view ray and the optimizer may accept implausible articulation for small pixel gains.

Single frame fitting with the L2 pose prior is the strongest in terms of pixel error on our data. The pose only variant is fast with about **0.91 s per frame** and **21.14 px** mean error. Enabling shape helps when person specific proportions matter and reduces the mean error to **20.49 px** while raising the runtime to **4.38 s per frame**. We compared the Avatar model GMM pose prior with the simple L2 prior and the table shows higher mean error for the GMM rows together with similar or longer time.

Qualitative inspection complements the numbers. In the turning pose with occlusions, the Avatar GMM prior often yields a mesh that twists less and shows fewer self intersections which is consistent with a stronger pose prior even though the pixel error is not lower on average. The multi frame method tends to preserve a consistent identity with stable shape and scale across time which is visible when the subject turns or moves toward the camera. At the same time it can smooth away fine per frame adjustments when temporal weight is high and this explains part of the gap in the table.

These observations align with the limits of monocular reconstruction in this setup. Absolute depth remains weakly constrained, keypoint coverage and noise dominate hard

frames, and our intrinsics are approximate which introduces small projection bias that the optimizer compensates with pose.



Figure 1. Qualitative comparison across fitting variants. Rows show two frames (“Neutral Pose” and “Turn Pose”). Each panel overlays the fitted SMPL mesh on the input image.

5. Conclusion

We presented a C++ pipeline that fits SMPL to monocular videos with Ceres and evaluated single-frame and prototype multi-frame variants. On our videos with a centered subject, single-frame fitting with an L2 pose prior is the most reliable in terms of pixel error. Pose-only gives the best speed and remains robust across clips, while adding shape helps when personalization matters at a moderate cost in runtime. The Avatar model GMM pose prior did not outperform the simple L2 prior in our setting and often led to slower solves with no 2D accuracy gain, but the 3D mesh followed more humanlike shapes. The multi-frame bundle adjustment with shared shape and temporal smoothness showed some qualitative benefits of steadier scale, but the current CPU implementation is very slow.

The study underlines the usual limits of monocular fitting. Depth and scale are weakly constrained, 2D keypoint noise and occlusions dominate hard frames, and approximate intrinsics introduce small systematic bias that the optimizer absorbs with pose.

6. Future Work

We aim to first strengthen the single-frame solver so it produces plausible poses even in difficult frames with occlusions or noisy keypoints. A learned pose prior such as VPoser could guide the optimizer toward human-like configurations that avoid self-intersections and excessive twisting. With a more dependable per-frame fit, the next focus is on the multi-frame setting: finding stable temporal weights, window length, and overlap, and adding light smoothing on translation and joint rotations, ideally with confidence weighting so weak frames do not pull strong ones.

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